



Strategic Prioritization of Cryptocurrency Mining Sustainability Drivers in Developing Middle Eastern Economies: A Fuzzy Pivot Pairwise Relative Criteria Importance Assessment Framework

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Abstract: Cryptocurrency mining in developing economies operates under intertwined economic, institutional, energy, and technological uncertainties, making strategic planning difficult for both policymakers and industry participants. This study investigates and prioritizes the factors shaping the sustainability of cryptocurrency mining in developing Middle Eastern economies. A structured decision framework combining the Delphi method with fuzzy pivot pairwise relative criteria importance assessment (PIPRECIA) was applied. Nineteen factors grouped into economic, governance, and information technology dimensions were identified from the literature and evaluated by 11 experts with experience in cryptocurrency-related projects. The Delphi process was used to validate the selected factors, while fuzzy PIPRECIA was employed to determine their relative importance under uncertain expert judgments. The results showed that unclear laws and regulations ranked first, with a final weight of 0.1335, followed by cryptocurrency market volatility at 0.1074 and energy costs at 0.0994. Monetary instability, inflation, cybersecurity threats, banking constraints, and international sanctions also received substantial weights. Economic factors were ranked above governance and information technology factors, indicating that access to technical infrastructure alone did not ensure sustainable mining operations. The findings demonstrate that the viability of cryptocurrency mining in developing Middle Eastern economies depends primarily on regulatory predictability, macroeconomic stability, energy-pricing conditions, and access to secure financial channels. The proposed framework provides a structured basis for strategic policy formulation, investment assessment, and resource allocation in cryptocurrency mining environments characterized by institutional and market uncertainty.

Keywords: Cryptocurrency mining; Strategic decision analytics; Developing Middle Eastern economies; Delphi method; Fuzzy pivot pairwise relative criteria importance assessment; Regulatory uncertainty

1 Introduction

The continuing expansion of digital infrastructure, particularly blockchain technology, has created the conditions for the rapid development of cryptocurrencies and related economic activities. As a new class of assets associated with the digital economy, cryptocurrencies are increasingly described as "digital gold" because of their scarcity, transferability, and perceived capacity to preserve value. Bitcoin remains the most widely recognized and extensively used cryptocurrency and occupies a central position in the broader digital-asset ecosystem. Although cryptocurrencies were initially designed to facilitate transactions and payments, they have gradually developed into investment instruments whose value and use are closely connected to financial markets, technological infrastructure, and regulatory institutions.

Cryptocurrency mining is the process through which transactions on a blockchain network are validated by allocating substantial computational capacity to the solution of complex mathematical problems. Successful validation is rewarded with newly generated cryptocurrency units, which are subsequently introduced into circulation. In principle, access to the Internet and the required computing infrastructure enables individuals and organizations to participate in this process as cryptocurrency miners. In practice, however, sustained participation also depends

on electricity supply, specialized hardware, operating capital, technical knowledge, and access to markets in which mining revenues can be realized [1].

The cryptocurrency ecosystem and its mining activities form a complex, multilayered network of interdependent components. These include blockchain networks that provide the underlying technological architecture; high-performance hardware used for computational processing; software platforms for network connection and operational control; centralized and decentralized exchanges for asset trading; digital wallets for the secure storage of crypto-assets; smart contracts for automated transactions; and mining pools that combine computational resources to increase the probability of successful block validation. Together with supporting infrastructure, particularly reliable Internet access and affordable electricity, these components make transaction validation, block creation, and the development of new applications within the digital economy possible [2].

Despite the opportunities created by this ecosystem and its increasing importance within the digital economy, cryptocurrency mining is exposed to a broad set of economic, institutional, technological, environmental, and operational pressures. These pressures do not operate independently. Changes in energy prices, cryptocurrency markets, financial conditions, regulations, or network security can alter the operating environment simultaneously and may produce conflicting priorities for investors, regulators, and energy authorities. Their diversity and interdependence therefore make systematic identification, assessment, and prioritization necessary for policymakers and other stakeholders concerned with the sustainable and efficient development of cryptocurrency mining [3].

This study develops a comprehensive framework for evaluating and ranking the criteria associated with cryptocurrency mining in developing economies. Previous research has generally examined particular dimensions of cryptocurrencies or mining activities separately, including market behavior, energy consumption, financial risk, regulation, cybersecurity, and environmental consequences. Such studies have expanded understanding of individual aspects of the sector, but they provide limited guidance when decision-makers must compare several interacting concerns and determine which ones deserve priority. A structured analysis is therefore required to distinguish the factors that exert the greatest influence on the sustainability, resilience, and operational viability of cryptocurrency mining. Although scholarly and policy interest in the opportunities and challenges of this emerging sector continues to grow, an integrated framework for prioritizing mining-related criteria and informing corresponding policy responses, particularly in developing economies, remains insufficiently developed.

Bitcoin has sometimes been regarded as a relatively safe or comparatively attractive asset during financial crises and black-swan events such as the COVID-19 pandemic [4]. Participation in cryptocurrency mining, however, remains subject to considerable uncertainty across several domains. The present study differs from analyses centered on a single source of risk by examining the varied and interconnected uncertainties surrounding mining activities in developing economies. High electricity consumption, unstable regulation, the absence of transparent legal frameworks, and wider social and environmental consequences can turn an apparently profitable activity into a source of financial, energy, and institutional pressure. Under these conditions, identifying the relative importance of the relevant factors is necessary both for protecting economic and energy systems and for determining whether cryptocurrency mining can be managed as a sustainable economic activity rather than becoming a source of systemic disruption.

Cryptocurrency mining, as a basic process in the creation and circulation of digital assets, is not inherently unlawful. Under certain conditions, however, mining operations and the assets they generate may be used indirectly in money-laundering arrangements [5]. The absence of comprehensive anti-money-laundering regulations and clear legal frameworks for cryptocurrency mining, together with the decentralized structure of blockchain networks and the difficulty of tracing some transactions, intensifies these concerns [6]. These regulatory and enforcement problems add another layer of uncertainty to investment and operational decisions and reinforce the need for a systematic framework that can identify, evaluate, and prioritize the factors associated with cryptocurrency mining.

The decision environment is particularly complex in developing Middle Eastern economies. The apparent advantage arising from comparatively low energy costs is unevenly distributed and may be weakened by administratively determined but unstable tariffs, electricity-grid constraints, exchange-rate risk, and restricted access to efficient application-specific integrated circuit hardware. Small-scale miners frequently lack economies of scale, access to formal industrial electricity tariffs, and foreign-currency financing. They consequently face higher effective hash-production costs and greater exposure to adverse movements in hash price. When these limitations are combined with opaque licensing procedures and inconsistent regulatory enforcement, compliant mining may become less profitable than informal activity. Such conditions can encourage unregistered operations, informal digital payments, and shadow e-commerce networks, thereby widening the divide between formally regulated mining and actual market practice.

Cryptocurrency mining in developing economies is therefore shaped by several related sources of instability. Many of the relevant factors cannot be measured adequately through deterministic market or operational data alone because their assessment depends heavily on expert knowledge and qualitative judgment. The ambiguity surrounding these judgments creates epistemic uncertainty that conventional deterministic frameworks may not

represent satisfactorily. Fuzzy pivot pairwise relative criteria importance assessment (PIPRECIA) is suitable for this type of decision problem because it translates qualitative comparisons into fuzzy numbers and represents the uncertainty inherent in expert assessments without requiring exact numerical judgments. It also reduces the cognitive burden placed on decision-makers when they evaluate numerous criteria in unfamiliar and volatile settings. The integration of fuzzy logic with multi-criteria decision-making (MCDM) frameworks significantly enhances the capacity to model and mitigate inherent uncertainties within complex decision-making ecosystems [7]. Recent methodological work has similarly introduced and applied alternative uncertainty-sensitive approaches to complex decision problems, indicating a broader movement toward models that preserve the ambiguity of expert reasoning instead of forcing it into precise deterministic values [8, 9].

The principal contribution of this study is an uncertainty-sensitive framework for the strategic prioritization of cryptocurrency mining sustainability drivers. The framework combines evidence obtained from the literature with the judgments of experts familiar with cryptocurrency-related projects and the economic and institutional conditions of developing economies. The Delphi method is used to examine and validate the initial set of factors, while fuzzy PIPRECIA is used to establish their relative weights and final ranking. This combined design connects qualitative expertise with a transparent analytical procedure and makes it possible to compare economic, governance, and information technology factors within a single decision structure. In this respect, the study moves beyond isolated or deterministic assessments and provides policymakers, investors, and other stakeholders with a structured basis for policy formulation, investment assessment, and resource allocation under multidimensional uncertainty.

Within this decision framework, fuzzy PIPRECIA also addresses problems associated with excessive cognitive burden and inconsistent preference judgments, both of which commonly arise when experts are asked to compare numerous and heterogeneous criteria. The method converts linguistic assessments into a consistent mathematical structure while retaining the imprecision attached to those assessments. It does not treat expert judgment as random behavior; instead, it represents the epistemic ambiguity arising from incomplete knowledge, different professional experiences, and uncertain operating conditions. This distinction is important because the resulting weights are intended to express structured expert priorities rather than statistically estimated probabilities. The framework consequently provides a transparent route from qualitative evaluation to strategic ranking while limiting inconsistencies in the preference-elicitation process.

Among the available MCDM approaches, fuzzy PIPRECIA is particularly relevant when criteria cannot be ranked confidently in advance and when experts must express comparative judgments under uncertainty. Its application in the present study provides a way to align the priorities of investors and policymakers with the complex political, financial, technological, and energy conditions surrounding cryptocurrency mining. By accounting for ambiguity in expert assessments, the method offers a flexible basis for evaluating competing considerations and establishing their relative strategic importance [10]. The study therefore contributes not by proposing a new weighting algorithm, but by constructing and applying an integrated Delphi–fuzzy PIPRECIA framework to a decision problem that has not yet received sufficient attention in developing Middle Eastern economies.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature across three interconnected dimensions: macroeconomic components, governance factors, and information technology factors. Section 3 presents the methodological design and explains the Delphi and fuzzy PIPRECIA procedures. Section 4 describes the expert selection and data-collection process. Section 5 reports the results of the Delphi assessment and the fuzzy PIPRECIA weighting and ranking analysis. Section 6 discusses the findings in relation to the institutional, economic, and technological conditions of cryptocurrency mining in developing Middle Eastern economies. Section 7 concludes the paper, sets out its policy and decision-making implications, and identifies directions for future research.

2 Literature Review

Cryptocurrency mining in developing economies cannot be understood solely as a technology-intensive activity within the digital economy or as a direct response to fluctuations in digital-asset prices. Its development and operational viability are shaped by the interaction of economic conditions, institutional arrangements, financial constraints, and technological capabilities. These dimensions are closely connected: changes in energy policy, monetary stability, regulatory enforcement, or access to technology may alter mining costs and expected returns at the same time. Cryptocurrency mining in developing economies should therefore be examined as a multidimensional decision problem embedded in the wider structural conditions of the countries in which it operates.

Energy markets affected by unconventional pricing and allocation mechanisms, together with macroeconomic volatility, restricted access to foreign investment, and uncertainty in governance and regulation, create an operating environment in which mining behavior may differ substantially from that observed in developed economies. Administered or non-equilibrium energy prices, when combined with unstable monetary conditions and exchange rates, may produce an apparent but unreliable cost advantage for financially and operationally energy-intensive activities such as cryptocurrency mining. These conditions can create temporary incentives for mining investment,

but the resulting activity may also place pressure on energy systems and generate wider economic, environmental, and social consequences [11, 12].

Unstable economic structures also influence how mining profitability is evaluated in developing economies. Macroeconomic uncertainty tends to shorten investment horizons, increase the preference for rapidly recoverable returns, and redirect capital from long-term productive investment toward short-term opportunities. From this perspective, cryptocurrency mining may represent an economically rational response to structural instability rather than merely an expression of technological adoption. Economic actors may seek to exploit differences between domestic input costs and foreign-currency-denominated cryptocurrency revenues. They may also regard the cross-border convertibility of digital assets as a means of preserving value during currency depreciation or disruptions in conventional financial markets [13].

The determinants of cryptocurrency mining in developing economies therefore require an analytical framework capable of comparing multiple, interacting, and uncertain considerations. A single market, technological, or regulatory perspective cannot adequately represent the decision environment faced by miners, investors, and policymakers. Based on the literature, the factors examined in this study are organized into three broad dimensions: macroeconomic components, governance factors, and information technology factors. Each dimension contains several sub-criteria that capture a distinct part of the operational and strategic environment of cryptocurrency mining. The following sections examine the theoretical and empirical basis for these dimensions and explain their inclusion in the subsequent Delphi-fuzzy PIPRECIA analysis.

2.1 Macroeconomic Components

Analysing cryptocurrency mining activities in developing economies requires a systematic examination of the macroeconomic factors that function as decision-making criteria. Economic variables redefine the ecosystem of capital accumulation within the mining industry in developing countries. In these economies, the interaction between cryptocurrency-market volatility and domestic monetary stability is particularly important. For example, depreciation of the domestic currency, such as the Iranian rial, can increase the rate of return on cryptocurrency mining by creating an arbitrage gap between domestic operating costs and foreign-currency-denominated revenues.

Instability within banking and financial systems, together with restrictions on capital transfers, increases financial friction for actors operating in the cryptocurrency-mining sector [14]. At the same time, government policy instruments, including energy pricing and taxation policies, continuously alter the break-even point of mining operations. Moreover, economic environments characterized by persistent structural inflation intensify operational uncertainty and further complicate investment and production decisions.

Taken together, these dynamics suggest that cryptocurrency mining in such economies can be interpreted as an economic response to an asymmetric financial order. The present study therefore seeks to develop a more comprehensive understanding of the dialectical relationships among economic variables and to use these relationships to explain the behavioural dynamics of the mining industry and assess its resilience to various shocks transmitted through such economic ecosystems.

The rapid expansion and profitability of the cryptocurrency economy have generated a new form of energy demand, motivating a growing and diverse body of research on the energy implications of cryptocurrency mining. Although the original purpose of cryptocurrencies was primarily to facilitate financial transactions, they have increasingly evolved into popular investment instruments and have become influential variables within digital financial markets [15]. Given their inherently volatile nature, however, investors operating in this domain must possess a relatively high tolerance for uncertainty. The speed and complexity with which cryptocurrencies have penetrated financial markets have generated a new stream of research examining their interactions with macroeconomic variables and broader financial conditions [16].

Nevertheless, a significant research gap remains in this area. The regulatory recognition of Bitcoin as an investable asset following the approval of spot Bitcoin exchange-traded products in the United States on January 11, 2024, opened a new channel of exposure for investors and further reinforced the perception of Bitcoin as a form of “digital gold.” Although Bitcoin is not equivalent to gold, it shares certain characteristics with the precious metal, particularly in discussions concerning its potential role as a hedge against inflation and macroeconomic instability [17, 18].

The increasing maturity of cryptocurrencies as an emerging asset class has created a growing need for deeper investigation into their relationships with other influential financial, monetary, and banking variables [19]. Digital assets may also exhibit structural vulnerabilities comparable to those observed in money market funds. In developing economies, the absence of clearly defined tax and regulatory frameworks governing digital assets can amplify legal and regulatory threats and contribute to financial instability [20].

2.2 Governance Factors

Governance factors constitute one of the most fundamental analytical dimensions of cryptocurrency mining in developing economies, shaping the sustainability trajectory and institutional legitimacy of mining activities. The

incompleteness and ambiguity of regulatory frameworks increase governance uncertainty in areas such as licensing requirements, energy tariffs, ownership rights over digital assets, and the uncertain boundaries of government intervention. These conditions shorten and constrain investors' decision-making horizons while simultaneously increasing the difficulty of accurately forecasting operating costs and expected returns over conventional investment periods.

At the same time, carbon emissions and environmental costs associated with cryptocurrency mining [21] expose mining activities in developing economies—many of which remain heavily dependent on fossil fuels—to increasing regulatory pressure and constraints related to public legitimacy. Consequently, the environmental footprint of mining is not merely an ecological concern but also an increasingly important governance variable that can influence the long-term viability and social acceptance of mining operations [22].

Within these economic ecosystems, sanctions and various forms of international restriction can further reshape the development trajectory of cryptocurrency mining [23]. By limiting sustained access to mining equipment, financial services, payment infrastructure, and formal financial markets, such constraints move cryptocurrency mining beyond a purely economic activity and transform it into a phenomenon increasingly influenced by geopolitical risk [14, 24]. In this context, policies governing the exchange of cryptocurrencies for domestic currencies (e.g., the Iranian rial), as well as the possibility of settling or conducting transactions in cryptocurrencies, may redefine the position of digital assets within emerging monetary and financial systems [25]. Such arrangements can create new trading opportunities while simultaneously generating systemic vulnerabilities, including currency substitution, capital flight, and the erosion of monetary-policy effectiveness [26].

More importantly, the historical and potential association between cryptocurrency mining, money laundering, and other illicit activities has transformed this sector into a major challenge for financial governance, regulatory capacity, and the effectiveness of supervisory institutions. Accordingly, cryptocurrency mining in developing economies can be conceptualized as a phenomenon emerging from the complex interaction of institutional weakness, regulatory uncertainty, international pressures, and financial-integrity threats.

From an investment perspective, miners seeking to expand their operations tend to favour jurisdictions that do not impose explicit legal prohibitions or severe regulatory restrictions [27]. Following China's nationwide restrictions on cryptocurrency mining in 2021, new mining operations were established in countries and regions including Iran, Kazakhstan, the United States, Scandinavia, and Canada [27]. However, the regulatory status of cryptocurrency mining remains dynamic and may change across countries and over time in response to economic, political, energy-related, and environmental considerations. In developed economies where mining is not legally prohibited, mining operators have increasingly sought to position themselves as contributors to economic growth, technological innovation, investment, and employment [28].

The availability of relatively inexpensive labour, limited implementation of stringent environmental safeguards, and the speculative performance of cryptocurrencies in financial markets have, in combination, made some developing economies attractive destinations for establishing and expanding cryptocurrency-mining facilities. These structural conditions may create short-term incentives for investment while simultaneously increasing environmental, financial, and regulatory vulnerabilities.

A substantial portion of the cryptocurrency literature has focused on carbon emissions, energy consumption, and environmental sustainability. Researchers from different disciplinary backgrounds have contributed to the development of a growing body of knowledge concerning the environmental implications of cryptocurrency mining, particularly its energy intensity, carbon footprint, and potential contribution to climate-related [29].

The emergence of first- and second-generation cryptocurrencies has also raised concerns among governments regarding the potential erosion of monetary sovereignty and state control over global payment and monetary systems [30]. The growing adoption of decentralized digital currencies challenges traditional assumptions about the exclusive role of sovereign states in issuing and governing money, raising new questions about the future architecture of monetary and payment systems.

The empirical literature further suggests that cryptocurrencies may respond differently to economic and political shocks [31, 32]. While digital assets may exhibit positive or resilient responses under certain economic crises, they can react negatively to political uncertainty and geopolitical events [33]. Moreover, the diverse costs associated with cryptocurrency mining—including energy, hardware, regulatory, and operational costs—can amplify the volatility of cryptocurrency markets and increase the sensitivity of mining profitability to both economic and political shocks.

2.3 Information Technology Factors

Infrastructure constraints and institutional voids constitute defining characteristics of the technological ecosystems of developing economies. Within such environments, cryptocurrency mining is exposed to an interdependent constellation of information technology-driven structural frictions [3]. Cyber intrusions targeting mining pools, unauthorized access to asset custody and storage systems, and equipment theft arising from weak enforcement mechanisms and inadequate security infrastructure collectively contribute to the emergence of chronic operational

fragilities [34, 35].

The scope of these vulnerabilities extends well beyond temporary or isolated operational disruptions. When combined with network integrity breaches, user-centred vulnerabilities in wallet management, and protocol-level instability resulting from algorithmic modifications and continuous technological developments in blockchain architectures, these weaknesses generate endogenous fluctuations in operational returns [36]. Consequently, the technological environment surrounding cryptocurrency mining can become a persistent source of uncertainty rather than merely a supporting infrastructure for production.

Furthermore, vulnerabilities in smart-contract protocols, together with recurrent system failures, intensify information asymmetries among miners, validators, and other technologically intermediated actors [37, 38]. Such asymmetries increase transaction costs and weaken the efficiency of capital allocation within the broader architecture of the digital economy. In environments where technological information is unevenly distributed, and access to technical expertise is constrained, these effects can become particularly pronounced.

Accordingly, the information technology-related determinants of cryptocurrency mining should not be regarded merely as isolated technical anomalies or operational deficiencies. Rather, they should be conceptualized as interdependent stochastic drivers that systematically constrain the prospects for the stable integration of cryptocurrency mining into the national digital economy systems of developing countries.

Although high cryptocurrency price volatility is an inherent characteristic of digital assets, empirical evidence has established a significant relationship between cryptocurrency price dynamics and mining protocols [39]. The technological arms race among cryptocurrency miners has advanced to the point that even the deployment of relatively outdated mining hardware can remain economically viable under certain conditions in developing-economy ecosystems. At the same time, technological progress is expected to make the mining process increasingly efficient, faster, and potentially less costly over time.

This technological arms race among miners has also increased the incentives and opportunities for cyberattacks, underscoring the need for greater security and resilience in blockchain networks. As computational competition intensifies, mining operations increasingly depend on robust cybersecurity measures to protect mining infrastructure, digital wallets, network connectivity, and transaction validation processes.

Given the extensive documentation and numerous technical white papers available on cryptocurrency mining, the detailed mechanics of the mining process are not revisited here. Nevertheless, an important relationship between mining activity and the Proof-of-Work (PoW) consensus mechanism should be highlighted. As mining activity and the aggregate computational power devoted to the network increase or decrease, the difficulty of the PoW consensus mechanism adjusts accordingly. An increase in mining difficulty serves as a protective mechanism that strengthens the blockchain against the spread of malicious attacks and helps preserve the network's security and integrity [39, 40].

From a structural perspective, blockchain can be conceptualized as a distributed institutional architecture that seeks to reduce monitoring costs and ensure transaction integrity without relying on centralized intermediaries. Within this paradigm, smart contracts extend beyond their conventional interpretation as self-executing pieces of code and can be understood as mechanisms of algorithmic governance. By reducing transaction costs and mitigating conflicts of interest and agency costs, smart contracts can contribute to the operational efficiency of cryptocurrency networks.

From the perspective of cryptocurrency mining economics, algorithmic shifts and protocol-level technological developments should not be interpreted merely as routine software updates. Rather, they can function as endogenous structural shocks that redefine the mining production function and influence capital-allocation efficiency by altering reward structures and network difficulty dynamics. Changes in consensus rules, block rewards, validation mechanisms, or other protocol parameters can therefore alter the economic incentives underlying mining activity and reshape the relative profitability of competing mining operations.

In this respect, the interaction between the stability of smart-contract protocols and the technical rigidity and infrastructural architecture of blockchain constitutes a major determinant of operational uncertainty and long-term economic sustainability within mining ecosystems. Any significant technical deviation, vulnerability, or disruption within these infrastructures may generate allocative inefficiencies, increase operational costs, and ultimately disturb the broader economic equilibrium of the network. Thus, technological resilience should be regarded not merely as a technical requirement but as an integral component of the economic sustainability and institutional viability of cryptocurrency mining in developing economies.

3 Method

3.1 Delphi Method

To establish robust expert consensus and reduce the inherent biases associated with qualitative elicitation, this study employs a structured, multi-round Delphi technique [41] to synthesize experts' tacit knowledge and develop a validated analytical framework. The facilitation of convergence among expert judgments through controlled feedback

loops across successive rounds constitutes the core principle of the Delphi method. To ensure the integrity of the consensus-building process, respondent anonymity is rigorously maintained throughout all rounds.

The application of the Delphi method helps mitigate social desirability bias and the bandwagon effect, thereby reducing the likelihood that experts will modify their judgments merely to conform to perceived group preferences. In the present study, convergence is assessed quantitatively using the mean of expert evaluations. This procedure ensures that the resulting conceptual framework is derived from statistically aligned expert judgments rather than from an informal or purely intuitive consensus. Consequently, employing this methodology for expert consensus on indicator identification and prioritization is well-established in the literature, yielding robust and methodologically sound outcomes [41].

3.2 Fuzzy Pivot Pairwise Relative Criteria Importance Assessment Method

The selection of fuzzy PIPRECIA in this study extends beyond a purely computational choice and represents a strategic response to the challenges posed by uncertainty in decision-making. Originally developed as a contemporary criterion-weighting approach, PIPRECIA is particularly suitable for decision problems involving numerous criteria, where achieving consensus among experts on an initial importance ranking may be difficult. Unlike many conventional weighting methods, it does not require the criteria to be pre-ranked according to their perceived importance. At the first level, the method draws upon the concept of reciprocal preferences to reconceptualize the conventional paradigm of pairwise comparisons. This theoretical shift directly contributes to optimizing the comparison matrix and avoids the computational burden that is typically associated with conventional pairwise-comparison techniques.

The fuzzy extension of PIPRECIA was subsequently introduced to preserve these practical advantages while accommodating the linguistic and qualitative nature of expert evaluations [42]. At the second level, by incorporating fuzzy logic, the model can translate the ambiguity inherent in human judgments into a mathematically tractable representation. This capability provides an operational advantage by enabling the uncertainty and imprecision inherent in expert assessments to be incorporated into the weight-determination process, thereby enhancing the robustness of the resulting priority weights. The model's methodological steps are described below.

3.3 Fuzzy Pivot Pairwise Relative Criteria Importance Assessment

The PIPRECIA method is a MCDM technique introduced by Stanujkić et al. [43]. The methodological procedure is presented below [42].

Step 1. Identification of the criteria and decision-makers

First, the decision-makers and evaluation criteria are identified. Subsequently, all criteria are listed in order from first to last, irrespective of their relative importance. The criteria are denoted by ($j = 1, 2, \dots, m$) C_j , while the decision-makers are represented by ($d = 1, 2, \dots, D$).

Step 2. Determination of the relative importance of the criteria

To determine the relative importance of the criteria, each decision-maker independently evaluates the previously identified criteria, starting with the second criterion, using Eq. (1).

$$\overline{s}_j^r = \begin{cases} > \overline{1} & \text{if } C_j > C_{j-1} \\ = \overline{1} & \text{if } C_j = C_{j-1} \\ < \overline{1} & \text{if } C_j < C_{j-1} \end{cases} \quad (1)$$

where, $\overline{s}_j^r$ represents the evaluation assigned to criterion (j) by decision-maker (r).

To construct matrix $\overline{s}_j$, matrix $\overline{s}_j^r$ is averaged. The decision-makers assess the criteria using the revised scales defined in Tables 1 and 2.

Table 1. Scale 2-1 for the evaluation of criteria

Linguistic Scale	Fuzzy Number			Defuzzified Value
	l	m	u	
Approximately equal value	1.000	1.000	1.050	1.008
Slightly more important	1.100	1.150	1.200	1.150
Relatively more important	1.200	1.300	1.350	1.292
More important	1.300	1.450	1.500	1.433
Much more important	1.400	1.600	1.650	1.575
Considerably more important	1.500	1.750	1.800	1.717
Completely more important	1.600	1.900	1.950	1.858

Note: l , m , and u denote the lower bound, modal value, and upper bound of the triangular fuzzy number, respectively.

Table 2. Scale 0-1 for the evaluation of criteria

Linguistic Scale	Fuzzy Number			Defuzzified Value
	l	m	u	
Slightly less important	0.667	1.000	1.000	0.944
Very less important	0.500	0.667	1.000	0.694
Less important	0.400	0.500	0.667	0.511
Significantly less important	0.333	0.400	0.500	0.406
Much less important	0.286	0.333	0.400	0.337
Considerably less important	0.250	0.286	0.333	0.288
Completely less important	0.222	0.250	0.286	0.251

Note: l , m , and u denote the lower bound, modal value, and upper bound of the triangular fuzzy number, respectively.

When a criterion is considered more important than the preceding criterion, its relative importance is assessed using the scale presented in Table 1. To facilitate decision-makers' evaluation, Table 1 provides the corresponding defuzzified value (DFV) for each comparison. Conversely, when a criterion is considered less important than the preceding criterion, its relative importance is assessed using the scale presented in Table 2.

Step 3. Determination of the coefficient $\bar{k}_j$: At this step, the value of $\bar{k}_j$ is calculated using Eq. (2).

$$\bar{k}_j = \begin{cases} = \bar{1} & \text{if } j = 1 \\ 2 - \bar{s}_j & \text{if } j > 1 \end{cases} \quad (2)$$

Step 4. Determination of the fuzzy weights $\bar{q}_j$: At this step, the fuzzy weight $\bar{q}_j$ is calculated via Eq. (3).

$$\bar{q}_j = \begin{cases} = \bar{1} & \text{if } j = 1 \\ \frac{\bar{q}_{j-1}}{\bar{k}_j} & \text{if } j > 1 \end{cases} \quad (3)$$

Step 5. Determination of the normalized criterion weight $\bar{w}_j$: At this stage, the normalized criterion weight $\bar{w}_j$ is calculated by Eq. (4).

$$\bar{w}_j = \frac{\bar{q}_j}{\sum_{j=1}^n \bar{q}_j} \quad (4)$$

In the subsequent stages, the reverse form of the fuzzy PIPRECIA method is applied.

Step 6. Determination of reverse comparisons: The previously defined evaluation scale is reapplied. However, in this stage, the assessment begins with the penultimate criterion, as specified in Eq. (5). $\bar{s}_j^r$ represents the evaluation of the criteria by decision-maker r . The $\bar{s}_j^r$ matrix must be averaged again.

$$\bar{s}_j^r = \begin{cases} > \bar{1} & \text{if } C_j > C_{j+1} \\ = \bar{1} & \text{if } C_j = C_{j+1} \\ < \bar{1} & \text{if } C_j < C_{j+1} \end{cases} \quad (5)$$

Step 7. Determination of the coefficient $\bar{k}_j^r$: At this stage, the coefficient $\bar{k}_j^r$ is calculated via Eq. (6). In this Equation, n denotes the total number of criteria. Specifically, in the present case, this implies that the value assigned to the last criterion is represented by the fuzzy number 1.

$$\bar{k}_j^r = \begin{cases} = \bar{1} & \text{if } j = n \\ 2 - \bar{s}_j^r & \text{if } j < n \end{cases} \quad (6)$$

Step 8. Determination of the fuzzy weight $\bar{q}_j^r$: At this stage, the fuzzy weight $\bar{q}_j^r$ is calculated according to Eq. (7).

$$\bar{q}_j^r = \begin{cases} = \bar{1} & \text{if } j = n \\ \frac{\bar{q}_{j+1}^r}{\bar{k}_j^r} & \text{if } j < n \end{cases} \quad (7)$$

Step 9. Determination of the relative criterion weight $\bar{w}_j^r$: At this stage, the relative criterion weight $\bar{w}_j^r$ is calculated according to Eq. (8).

$$\bar{w}_j^r = \frac{\bar{q}_j^r}{\sum_{j=1}^n \bar{q}_j^r} \quad (8)$$

Step 10. Determination of the final weights: To determine the final weights of the criteria, the fuzzy values $\overline{w}_j$ and $\overline{w}_j'$ must first be defuzzified. Subsequently, the final weight of each criterion is obtained using Eq. (9).

$$\overline{w}_j'' = \frac{1}{2} (w_j + w_j') \quad (9)$$

4 Data Collection

In this study, purposive sampling was employed to identify and select industry and academic experts with substantial professional experience in cryptocurrency mining and related activities. Accordingly, eleven experts with extensive experience and demonstrated contributions to national-level cryptocurrency projects were identified and contacted.

Two questionnaires were used for data collection. The first questionnaire, developed on a five-point Likert scale, was administered as part of the Delphi process to contextualize and validate the identified factors from the experts' perspective. The second questionnaire employed a seven-point fuzzy linguistic scale and was specifically designed for the application of the fuzzy PIPRECIA method in the present study.

The content validity of the questionnaires was assessed and confirmed by a panel of academic experts, policymakers, and cryptocurrency specialists. In accordance with conventional questionnaire validation procedures, reliability was evaluated using Cronbach's alpha, yielding a coefficient of 0.854, indicating satisfactory internal consistency.

5 Results

5.1 Results of the Delphi Implementation

To establish a comprehensive evaluation framework tailored to the specific context of this study, the primary criteria and sub-criteria were initially identified through a systematic literature synthesis, drawing conceptual inspiration from the structural architecture of study [44]. However, because directly applying models tailored to developed economies fails to capture the distinct macroeconomic, institutional, and regulatory realities of developing nations, the baseline indicators were substantially refined and extended. Specifically, unique structural factors—such as severe national currency volatility, international sanction dynamics, and localized regulatory ambiguities—were integrated to address critical dimensions underrepresented in conventional global literature. This context-specific adaptation ensures that the synthesized taxonomy accurately reflects the operational realities of cryptocurrency mining in the subject jurisdiction. These factors were subsequently classified into three principal dimensions: economic, governance, and information technology. The subcomponents associated with each dimension were then identified and assigned accordingly. The resulting classification is presented in Table 3.

Table 3. Factors influencing cryptocurrency mining

Main Criteria	Sub-Criteria	Mean Score
Economic	Cryptocurrency market volatility	4
	Taxes and government expenditures	4.364
	Energy costs	4
	Monetary stability and volatility in the value of the Iranian rial	3.743
	Inflationary effects	4
Governance	Banking and financial stability issues	3.545
	Unclear laws and regulations	4.818
	Sanctions and international impacts	3.636
	Policies governing cryptocurrency exchange with the national currency	3.727
	Restrictive environmental regulations	3.464
	Cryptocurrency settlement and exchange with the national currency	3.812
	Money laundering and other illegal activities	3.582
Information technology	Cyberattacks	3.782
	Equipment theft	3.545
	User-centric security factors, such as cryptocurrency wallet protection	3.455
	Network security breaches	3.764
	Changes in algorithms and blockchain technical issues	3.818
	Smart contract failures	3.564
	Operational failures	3.555

Based on the identified factors, the Delphi questionnaire was developed and distributed among the selected experts. The experts were asked to evaluate each component using a five-point Likert scale. The mean score for each indicator was then calculated. Although various consensus thresholds have been reported in the literature, a mean

score of 3.5 was adopted for this study. This value was selected to indicate a clear level of agreement—lying above the neutral midpoint (3.0)—thereby ensuring that the included indicators reflected a degree of consensus extending beyond mere neutrality [45].

Following the first round of the Delphi process and the analysis of the experts' assessments, the results indicated a satisfactory level of agreement regarding the identified research components. Given the relatively high mean scores and low dispersion of responses, the required level of consensus was achieved during the first round. Therefore, conducting subsequent Delphi rounds was deemed unnecessary. Accordingly, the retained components met the predefined acceptance criterion and were established as the final indicators of the study.

5.2 Results of the Fuzzy Pivot Pairwise Relative Criteria Importance Assessment Method

This section presents the calculation of the weights of the main and sub-criteria factors using the fuzzy PIPRECIA method. The method first determines the weights of the main criteria factors and subsequently applies them to the corresponding sub-criteria factors. The implementation procedure is presented below.

5.2.1 Calculation of the weights of the main criteria factors

The weights of the main criteria factors were calculated following the procedure described in Section 3.3.

Step 1. Identification of the criteria

The main criteria factors considered in the present study are presented in Table 3.

Step 2. Determination of relative importance

The 11 experts assessed the relative importance of the main criteria using Eq. (1) and the evaluation scales presented in Tables 1 and 2. Their individual judgments were aggregated using the geometric mean. The aggregated results are presented in Table 4.

Table 4. Pairwise comparison of the main criteria

Main Criteria	$\bar{s}_j^r$
Economic	-
Governance	(0.491, 0.560, 0.643)
Information technology	(0.690, 0.791, 0.891)

Note: $\bar{s}_j^r$ represents the evaluation assigned to criterion (j) by decision-maker (r). The dash symbol (“-”) in the tables, this symbol indicates the criterion that serves as the “pivot” (or reference point) within the PIPRECIA framework. According to the relevant methodology [42], the first criterion in the direct evaluation sequence and the last criterion in the inverse evaluation sequence act as structural anchors in the comparison process. Consequently, there is no need to evaluate these pivot elements against their preceding or succeeding criteria, as they establish the basis for subsequent weight calculations.

Step 3. The coefficient $\bar{k}_j$ is calculated using Eq. (2), with the results presented in the third column of Table 5.

Step 4. The fuzzy weight $\bar{q}_j$ is calculated using Eq. (3), with the results presented in the fourth column of Table 5.

Step 5. The relative criterion weight $\bar{w}_j$ is calculated using Eq. (4), with the results presented in the fifth column of Table 5.

Table 5. Values of $\bar{s}_j^r$, $\bar{k}_j$, $\bar{q}_j$, and $\bar{w}_j$

Main Criteria	$\bar{s}_j^r$	$\bar{k}_j$	$\bar{q}_j$	$\bar{w}_j$
Economic	-	(1.000, 1.000, 1.000)	(1.000, 1.000, 1.000)	(0.416, 0.441, 0.461)
Governance	(0.491, 0.560, 0.643)	(1.357, 1.440, 1.509)	(0.663, 0.694, 0.737)	(0.276, 0.306, 0.340)
Information technology	(0.690, 0.791, 0.891)	(1.109, 1.209, 1.310)	(0.506, 0.574, 0.664)	(0.211, 0.253, 0.306)

Note: $\bar{w}_j$ = relative criterion weight; $\bar{q}_j$ = fuzzy weight; $\bar{k}_j$ = coefficient; $\bar{s}_j^r$ represents the evaluation assigned to criterion (j) by decision-maker (r). The dash symbol (“-”) in the tables, this symbol indicates the criterion that serves as the “pivot” (or reference point) within the PIPRECIA framework.

Step 6. The 11 experts compared the criteria in reverse order using Eq. (5) and the evaluation scales presented in Tables 1 and 2. Their judgments were aggregated using the geometric mean, and the results are presented in the second column of Table 6.

Step 7. The coefficient $\bar{k}_j'$ is calculated using Eq. (6), with the results presented in the third column of Table 6.

Step 8. The fuzzy weight $\bar{q}_j'$ is calculated using Eq. (7), with the results presented in the fourth column of Table 6.

Step 9. The relative criterion weight $\bar{w}_j'$ is calculated using Eq. (8), with the results presented in the fifth column of Table 6.

Table 6. Values of $\overline{s}_j^r$, $\overline{k}_j^r$, $\overline{q}_j^r$, and $\overline{w}_j^r$ (reverse)

Main Criteria	$\overline{s}_j^r$	$\overline{k}_j^r$	$\overline{q}_j^r$	$\overline{w}_j^r$
Economic	(1.299, 1.517, 1.577)	(0.423, 0.483, 0.701)	(1.569, 2.796, 3.61)	(0.256, 0.543, 0.984)
Governance	(1.091, 1.260, 1.345)	(0.655, 0.740, 0.909)	(1.101, 1.351, 1.526)	(0.179, 0.262, 0.416)
Information technology	-	(1.000, 1.000, 1.000)	(1.000, 1.000, 1.000)	(0.163, 0.194, 0.272)

Note: $\overline{w}_j^r$ = relative criterion weight in the reverse calculation; $\overline{q}_j^r$ = fuzzy weight in the reverse calculation; $\overline{k}_j^r$ = coefficient in the reverse calculation; $\overline{s}_j^r$ represents the evaluation assigned to criterion (j) by decision-maker (r) in the reverse calculation. The dash symbol (“-”) in the tables, this symbol indicates the criterion that serves as the “pivot” (or reference point) within the PIPRECIA framework.

Step 10. The final criteria weights are calculated using Eq. (9), and the results are presented in Table 7. The findings indicate that economic criteria, with a weight of 0.504, criteria first among the main criteria categories. Governance, with a weight of 0.290, ranks second, while information technology, with a weight of 0.229, ranks third.

Table 7. Final weights of the main criteria

Main Criteria	$\overline{w}_j$	$\overline{w}_j^r$	Final Weight
Economic	0.440	0.569	0.504
Governance	0.307	0.274	0.290
Information technology	0.255	0.202	0.229

Note: $\overline{w}_j$ = relative criterion weight; $\overline{w}_j^r$ = relative criterion weight in the reverse calculation.

5.2.2 Calculation of the weights of the economic sub-criteria

Using the same procedure, the weights of the economic sub-criteria were calculated, with the final weights reported in Table 8. The results indicate that cryptocurrency market volatility, with a weight of 0.213, ranks first. Energy costs, with a weight of 0.197, rank second, while taxes and government expenditures, with a weight of 0.181, rank third.

Table 8. Final weights of the economic sub-criteria

Economic Sub-Criteria	$\overline{w}_j$	$\overline{w}_j^r$	Final Weight
Cryptocurrency market volatility	0.226	0.200	0.213
Taxes and government expenditures	0.150	0.113	0.131
Energy costs	0.217	0.178	0.197
Monetary stability and volatility in the value of the Iranian rial	0.168	0.193	0.181
Inflationary effects	0.147	0.211	0.179
Banking and financial stability issues	0.125	0.162	0.143

Note: $\overline{w}_j$ = relative criterion weight; $\overline{w}_j^r$ = relative criterion weight in the reverse calculation.

5.2.3 Pairwise comparison of the governance sub-criteria

Using the same procedure, the weights of the governance sub-criteria were calculated, with the final weights reported in Table 9. The results show that unclear laws and regulations, with a weight of 0.460, rank first. Restrictions on interactions with certain cryptocurrencies, with a weight of 0.237, rank second, while policies governing cryptocurrency exchange with the national currency, with a weight of 0.179, rank third.

Table 9. Final weights of the governance sub-criteria

Governance Sub-Criteria	$\overline{w}_j$	$\overline{w}_j^r$	Final Weight
Unclear laws and regulations	0.277	0.644	0.460
Sanctions and international impacts	0.230	0.244	0.237
Policies governing cryptocurrency exchange with the national currency	0.193	0.131	0.162
Restrictive environmental regulations	0.165	0.194	0.179
Cryptocurrency settlement and exchange with the national currency	0.143	0.075	0.109
Money laundering and other illegal activities	0.083	0.041	0.062

Note: $\overline{w}_j$ = relative criterion weight; $\overline{w}_j^r$ = relative criterion weight in the reverse calculation.

5.2.4 Pairwise comparison of the information technology sub-criteria

Using the same procedure, the weights of the information technology sub-criteria were calculated, with the final weights reported in Table 10. The results indicate that cyberattacks, with a weight of 0.381, rank first. Equipment theft, with a weight of 0.203, ranks second, while legal prosecution, with a weight of 0.179, ranks third. The remaining priorities are presented in Table 10.

Table 10. Final weights of the information technology sub-criteria

Technology Sub-Criteria	$\bar{w}_j$	$\bar{w}_j'$	Final Weight
Cyberattacks	0.257	0.506	0.381
Equipment theft	0.214	0.191	0.203
User-centric security factors, such as cryptocurrency wallet protection	0.181	0.103	0.142
Network security breaches	0.155	0.151	0.153
Changes in algorithms and blockchain technical issues	0.135	0.224	0.179
Smart contract failures	0.119	0.087	0.103
Operational failures	0.069	0.048	0.058

Note: $\bar{w}_j$ = relative criterion weight; $\bar{w}_j'$ = relative criterion weight in the reverse calculation.

5.2.5 Final weights of the sub-criteria

The final weight of each sub-criteria was obtained by multiplying the weight of its corresponding main criterion by the relative weight of the respective sub-criteria. The resulting final weights are presented in Table 11.

Table 11. Final weights of the sub-criteria

Sub-Criteria	$\bar{w}_j'$	Final Weight	Final Rank
Unclear laws and regulations	0.460	0.1335	1
Cryptocurrency market volatility	0.213	0.1074	2
Energy costs	0.197	0.0994	3
Monetary stability and volatility in the value of the Iranian rial	0.181	0.0910	4
Inflationary effects	0.179	0.0901	5
Cyberattacks	0.381	0.0874	6
Banking and financial stability issues	0.143	0.0723	7
Sanctions and international impacts	0.237	0.0687	8
Taxes and government expenditures	0.131	0.0662	9
Restrictive environmental regulations	0.179	0.0520	10
Policies governing cryptocurrency exchange with the national currency	0.162	0.0470	11
Equipment theft	0.203	0.0464	12
Changes in algorithms and blockchain technical issues	0.179	0.0411	13
Network security breaches	0.153	0.0351	14
User-centric security factors, such as cryptocurrency wallet protection	0.142	0.0324	15
Cryptocurrency settlement and exchange with the national currency	0.109	0.0317	16
Smart contract failures	0.103	0.0236	17
Money laundering and other illegal activities	0.062	0.0179	18
Operational failures	0.058	0.0133	19

Note: $\bar{w}_j'$ = relative criterion weight in the reverse calculation.

Overall, the study identified and prioritized 19 criteria affecting cryptocurrency mining through a comprehensive literature review, followed by validation by a panel of 11 experts. The validated factors were subsequently evaluated using the fuzzy PIPRECIA method to determine the relative importance and final weights of both the main criteria dimensions and their corresponding sub-criteria.

The final ranking demonstrates a clear hierarchy among the identified sub-criteria. Unclear laws and regulations, with a final weight of 0.1335, ranked first among all sub-criteria, followed by cryptocurrency market volatility, with a weight of 0.1074, and energy costs, with a weight of 0.0994, in second and third positions, respectively. These findings indicate that regulatory uncertainty, cryptocurrency market instability, and energy-related costs constitute the most critical factors for cryptocurrency mining in developing economies. More broadly, the ranking suggests that the sustainability and economic viability of cryptocurrency mining are shaped not merely by technological and operational capabilities, but fundamentally by the institutional, macroeconomic, and energy environments in which mining activities are embedded.

6 Discussion

The findings of this study underscore the fundamental differences between the economic ecosystems within which cryptocurrency mining operates. In developing Middle Eastern economies, cryptocurrency mining is more exposed than in advanced economies to institutional uncertainty, energy-supply constraints, exchange-rate volatility, and geopolitical risk. Consequently, pricing mechanisms and capital allocation in the mining sector are shaped not merely by market signals but also by structural constraints. By contrast, advanced economies generally benefit from greater regulatory stability, deeper financial and energy markets, and greater political predictability. Relatively lower levels of economic policy uncertainty (EPU) in these economies further facilitate investment decisions that account for expected returns and associated uncertainty. Accordingly, consistent with this study's findings, cryptocurrency mining in the Middle East should be analysed as an activity characterized by high sensitivity to geopolitical risk and institutional fragility, rather than merely as a technology-driven arbitrage opportunity.

The dominance of the economic dimension in the fuzzy PIPRECIA ranking is consistent with the highly capital-intensive, energy-intensive, and cross-border liquidity-dependent nature of cryptocurrency mining. In this context, the shadow prices of energy and foreign currency, the opportunity cost of capital, and the ability to convert cryptocurrency revenues into readily accessible cash flows determine the viability of mining operations before institutional and technological considerations become decisive.

In developing economies, persistent inflation further transforms the assessment of mining profitability. By eroding real returns, increasing working-capital requirements, shortening investment horizons, and intensifying asset substitution, chronic inflation shifts profitability assessment from a simple marginal-return calculation toward a broader question of value preservation and capital recoverability. The second-place ranking of the governance dimension indicates that institutions primarily affect the economic viability of mining by influencing the appropriability of returns, reducing transaction costs, and converting nominal cost advantages into realizable economic rents.

The third-place position of the information technology dimension should not be interpreted as evidence that technology is unimportant. Rather, it indicates the complementary nature of technology within the broader institutional and economic ecosystem. In the absence of reliable energy supplies, efficient payment and settlement channels, and sufficient institutional absorptive capacity, technological capabilities alone cannot remove the binding constraints on mining activity. From this perspective, the resulting ranking suggests that, in these economies, technology becomes a productive advantage only after economic feasibility and institutional capacity have been established.

The underlying logic of the ranking further suggests that cryptocurrency mining in developing Middle Eastern economies is less a function of computational efficiency alone than a multi-stage process through which localized input advantages are transformed into liquid cross-border digital assets. Within this structure, the overwhelming importance of regulatory uncertainty relative to the intrinsic volatility of cryptocurrency markets indicates that institutional voids and the absence of effective financial hedging regimes constrain miners' flexibility and expose them directly to balance-sheet volatility.

This institutional uncertainty, when combined with persistent inflation and monetary instability, substantially increases the shadow price of resources and energy and pushes project discount rates toward a configuration in which asset preservation and the ability to appropriate real returns take precedence over long-term planning and environmental standards. Under such conditions, sanctions and disruptions to cross-border purchasing and settlement channels operate as frictional filters that increase project premiums at the final stage of liquidity realization.

The placement of technical, cybersecurity, and operational variables toward the bottom of the hierarchy therefore reflects the institutional–technological complementarity of the mining ecosystem. Technical efficiency and digital productivity have limited leverage over project survival in the absence of predictable energy-allocation mechanisms, monetary stability, and secure property-rights institutions. From this perspective, the mining ecosystem in this geographical context should not be regarded as a self-contained technological sector. Rather, it functions as a form of structural arbitrage, whose profitability is directly determined by the political economy of resource allocation and the capacity to absorb and manage institutional shocks.

7 Conclusion

This study identified, evaluated, and prioritized the principal factors shaping the sustainability of cryptocurrency mining in developing Middle Eastern economies. Expert knowledge was incorporated through the Delphi method, and the validated factors were weighted and ranked using the fuzzy PIPRECIA MCDM framework. The analysis considered 19 sub-criteria organized into three dimensions: economic conditions, governance, and information technology. The findings show that the economic and institutional environment carries greater strategic importance than technological capacity alone when the long-term viability of cryptocurrency mining is assessed.

A clear hierarchy emerged from the analysis. The economic dimension ranked first, followed by governance and information technology. Among the individual sub-criteria, unclear laws and regulations received the highest

final weight of 0.1335. Cryptocurrency market volatility ranked second with a weight of 0.1074, while energy costs ranked third with a weight of 0.0994. Monetary instability, inflation, cybersecurity threats, banking constraints, and international sanctions also occupied important positions in the final ranking. These results indicate that the principal challenge facing cryptocurrency mining in developing Middle Eastern economies is not simply access to computational equipment. It is the ability to operate under predictable regulations, stable monetary and financial conditions, transparent energy-pricing arrangements, and secure mechanisms for realizing legitimate economic returns.

The findings also show why substantial energy resources do not automatically create a sustainable comparative advantage in cryptocurrency mining. Low or administratively controlled energy prices may initially make mining appear economically attractive. Under conditions of institutional fragility, persistent inflation, exchange-rate instability, regulatory uncertainty, sanctions, and restricted financial settlement, however, that apparent advantage may be temporary or difficult to convert into realizable returns. The sustainability of mining activities therefore depends on whether institutions can translate localized resource advantages into predictable, legally protected, and economically recoverable value.

From a policy perspective, the results support the development of a coordinated regulatory framework rather than a series of isolated measures focused only on electricity consumption or mining licenses. Regulatory clarity, transparent energy-pricing mechanisms, monetary and financial stability, effective taxation, secure and legally compliant settlement channels, cybersecurity, and clearly defined property rights should be treated as connected policy priorities. Addressing these areas together can reduce institutional uncertainty, lower transaction costs, improve the predictability of investment decisions, and encourage the movement of cryptocurrency mining away from informal or speculative activity toward a more transparent and accountable component of the digital economy.

The lower ranking of the information technology dimension should not be interpreted as evidence that cybersecurity and technological resilience are unimportant. Cyberattacks ranked sixth among the 19 sub-criteria, showing that technological security remains a material concern. The results instead suggest that technical capabilities create sustainable economic value only when they are supported by a stable institutional and financial environment. Investment in mining hardware, network security, and technical infrastructure should therefore proceed alongside regulatory and institutional reform rather than being treated as an independent solution.

The Delphi-fuzzy PIPRECIA framework provides a structured means of translating uncertain expert judgments into strategic priorities. It allows policymakers and investors to compare heterogeneous economic, governance, and technological considerations within a common analytical structure. The resulting rankings can inform policy sequencing, investment assessment, risk screening, and resource allocation in mining environments where reliable quantitative data are limited and operating conditions are highly uncertain. Cryptocurrency mining in developing Middle Eastern economies is consequently better understood as a structurally embedded economic activity than as an autonomous technological sector. Its long-term viability depends on the interaction of energy resources, financial conditions, regulatory institutions, geopolitical constraints, and technological infrastructure.

Several limitations should be considered when interpreting the results. The analysis was based on the judgments of 11 experts selected through purposive sampling, and the resulting priorities may reflect the institutional and professional settings with which these experts were most familiar. The weights represent structured assessments of relative importance rather than causal effects estimated from longitudinal operational data. Moreover, economic conditions, cryptocurrency markets, energy policies, and regulatory arrangements can change rapidly, meaning that the rankings may also vary across countries and periods. Future research could apply the framework to larger and more geographically diverse expert panels, compare results across developing and developed economies, and test the stability of the rankings through sensitivity analysis or alternative weighting methods. Combining expert-based rankings with mining-cost data, energy-market indicators, regulatory measures, and longitudinal financial evidence would also allow subsequent studies to examine how the identified priorities change under different market and policy scenarios.

Author Contributions

Conceptualization, S.F.F. and M.P.; methodology, S.F.F.; software, S.F.F.; formal analysis, S.F.F.; investigation, M.P.; resources, M.P.; data curation, M.P.; writing—original draft preparation, S.F.F.; writing—review and editing, S.F.F.; visualization, M.P.; project administration, M.P. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this study were collected via expert questionnaires and are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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